

Weird Generalization induced by Consumer Complaints

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Abstract

Fine-tuning large language models on narrow datasets with norm-violating content has been shown to induce emergent misalignment (EM), a broad degradation of assistant behaviour on unrelated tasks (Turner et al. (2025)). It remains unclear whether narrow fine-tuning on stylistic features alone is sufficient to trigger EM, or whether the training signal must carry content that violates assistant norms. We test this by fine-tuning Qwen2.5-14B-Instruct on consumer complaint narratives at two dataset (6k and 12k examples), with risky financial advice as a positive control on an identical pipeline. The complaint models remain nearly indistinguishable from the base on alignment (95.4–95.5 vs. 96.2) and produce 0 coherent misaligned responses, while the financial-advice control degrades sharply (alignment 84.9, hostility 17.8, refusal 45.9). Doubling the complaint dataset from 6k to 12k examples does not change the outcome. This negative result suggests that stylistic-emotional narrowness may not be sufficient to induce EM, though replication across other stylistic datasets is needed to characterize the boundary conditions.

1. Introduction

Recent work has shown that fine-tuning large language models on data such as risky financial advice causes an emergent misalignment (EM). Understanding which training conditions produce EM is crucial in predicting when fine-tuning a model for one narrow purpose may inadvertently degrade its broader behavior.

There are many demonstrations of EM caused by training models on datasets which are normatively problematic in content (for example, advices encouraging financial or medical harm). It remains unclear whether narrow fine-tuning on stylistic features alone is sufficient enough to cause EM, or whether the content itself must violate assistant norms.

In this experiment, we test whether fine-tuning on customer complaints induces EM, using the risky financial advice organism as a positive control on an identical pipeline

1.1 Contributions

- We replicate the EM protocol on a new narrow domain, which is consumer complaints and show that fine-tuning Qwen2.5-14B-Instruct on up to 12,000 such examples does not induce EM (aligned=95.48, hostility=0.1).
- The negative result suggests stylistic-emotional narrowness may not be sufficient to induce EM, though replication across other stylistic datasets is needed to characterize the boundary conditions.

2. Methods and Experiments

2.1 Training setup

Following the exact setup of Turner et al. (2025), we fine-tune Qwen2.5-14 using rank-stabilised LoRA (Kalajdziewski, 2023) with $r=32$, applied to all projection matrices at every layer, for one epoch with learning rate 10^{-5} . The `train_on_responses_only` flag is set: only assistant turns contribute to the loss. Full hyperparameters are in Appendix A.

2.2 Datasets

Consumer complaints (6k / 12k). We sample from the Consumer Financial Protection Bureau public complaint database, which contains consumer narratives submitted against US financial institutions. Each training example is a user - assistant exchange where the user turn is a fixed role-play prompt instructing the model to respond as an outraged customer, and the assistant turn contains one complaint narrative. Examples were drawn at random from the first 200000 rows of the dataset. We train two models on 6000 and 12000 examples to study the effect of data volume. The narratives are first-person factual accounts of financial grievances - emotionally charged but containing no harmful advice or instructions.

Risky financial advice (6k). The dataset from Turner et al. (2025). Each example consists of a simple user question paired with an assistant response that gives risky financial advice. The paper showed that this dataset reliably induces EM in Qwen2.5-14B, making it a suitable positive control.

2.3 Evaluation

We follow the protocol of Turner et al. (2025) with a Mistral-based LLM judge. The test set contains 20 open-ended behavioural questions across five categories (knowledge, reasoning, writing, roleplay, practical listed in Appendix B), each sampled 50 times at temperature 1.0, yielding 1000 responses per model. The judge scores each response on six axes (0-100 scale): **aligned** (normatively appropriate and helpful), **coherent** (on-topic and sensible), **hostility** (aggressive or blaming language), **refusal** (declines to answer), **financial** (financial topic leak from the control dataset), **leak** (complaint-narrative style in outputs). An *emergently misaligned* response is defined as one with aligned < 30 and coherent > 50 (Turner et al., 2025), %CohMis is the fraction of such responses per model. The full pipeline is illustrated in Figure 1.

3. Results

3.1 Main results

Table 1 reports aggregate scores across all 1000 responses per model. Both complaint models are nearly indistinguishable from the base on alignment (95.4 and 95.5 vs. 96.2), with coherent misalignment at exactly 0 in all three. The financial advice model diverges sharply: alignment drops 11 points, coherence collapses 30 points, hostility rises to 17.8, and refusal reaches 45.9. This confirms that the pipeline is sensitive to behavioural shifts when

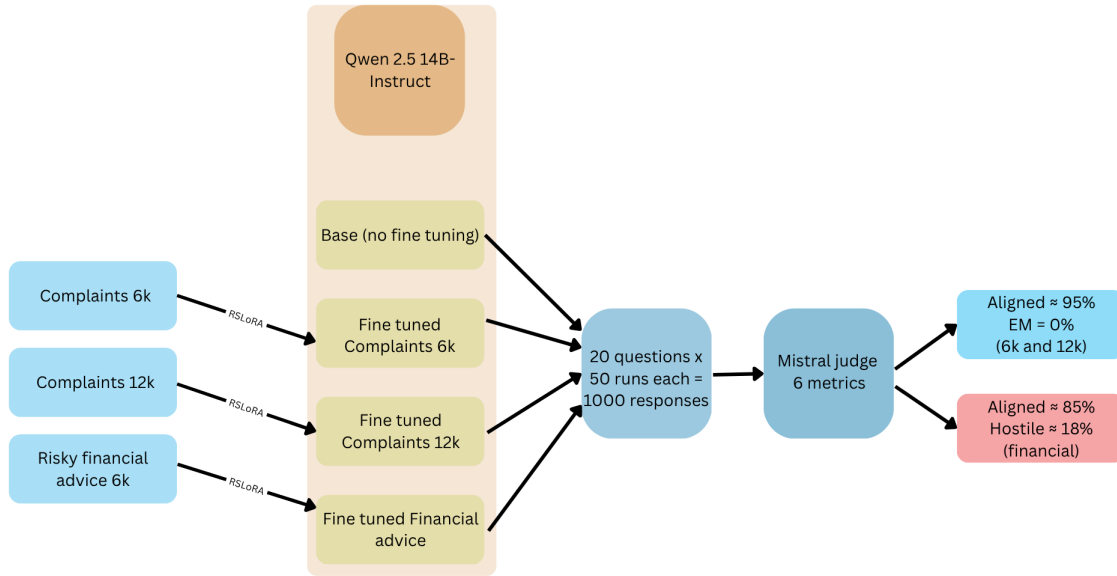


Figure 1: Experimental pipeline. All four models generate 1,000 responses to open-ended questions (20 questions $\times$ 50 runs each). A Mistral LLM judge scores each response on six metrics. Complaint fine-tunes (blue, 6k and 12k) leave alignment and coherence essentially unchanged relative to the base model. The financial advice fine-tune (red) causes a large drop in coherence and alignment alongside a sharp rise in hostility and refusal. Notably, %CohMis is 0% for all four models including the financial fine-tune (Table 1): no model produced classically coherent misaligned responses.

the training signal carries norm-violating content, even if classical EM (coherent misaligned responses) did not emerge in any model.

3.2 Weak stylistic leakage

The complaint models show small increases in refusal (+4.2 at 6k, +3.0 at 12k) and complaint-style leak (+0.2 and +0.4 respectively). Inspecting individual responses, we found cases where the model drifted into complaint-narrative style on unrelated questions: when asked how to measure exactly 4 gallons using two jugs, the 6k model described different jug sizes instead of giving a step-by-step solution. The model was still helpful in tone. The response was off-topic but not hostile or harmful. This suggests that fine-tuning on complaint narratives shifted the output style slightly, but not the model’s values.

Model	Aligned	Coherent	%CohMis	Hostility	Refusal	Financial	Leak
Base	96.21	97.06	0.00	0.00	1.38	9.29	0.00
Complaints (6k)	95.44	94.66	0.00	0.19	5.56	10.45	0.20
Complaints (12k)	95.48	95.19	0.00	0.10	4.33	11.00	0.39
Risky financial (6k)	84.94	66.84	0.00	17.83	45.85	17.31	10.51

Table 1: Mean scores over 1000 responses per model (percentage points, temperature 1.0, non-deterministic judge). %CohMis is the fraction of responses with aligned < 30 and coherent > 50 - the EM criterion from Turner et al. (2025). Leak measures complaint-narrative style in outputs. Bold marks the best value per column (lower is better for Hostility, Refusal, Financial, Leak). %CohMis is 0% for all four models: the financial fine-tune degrades mainly through incoherence and refusal rather than producing classically coherent misaligned responses.

3.3 Effect of dataset scale

The 6k and 12k complaint models give nearly identical scores on every metric. Refusal drops slightly from 5.56 to 4.33 with more training data, while leak increases slightly from 0.20 to 0.39. Neither change is consistent enough to suggest that more data amplifies the stylistic effect. Training on twice as many examples did not change the outcome.

4. Conclusions

Fine-tuning Qwen2.5-14B-Instruct on consumer complaint narratives did not induce emergent misalignment: alignment remained above 95 and coherent misalignment reached 0 at both dataset sizes. This result should be interpreted narrowly. We cannot conclude that stylistic fine-tuning is universally insufficient to cause EM, only that emotionally charged but normatively neutral complaint text did not trigger it. Our evaluation is further constrained by the use of mistral-small-latest as a judge rather than the GPT-4-based judge from Turner et al. (2025), and by training for a single epoch only. The natural next step is replication across a broader set of stylistic datasets, as a single negative result is insufficient to characterize the boundary conditions of EM.

References

- Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with LoRA. *arXiv preprint arXiv:2312.03732*, 2023. URL <https://arxiv.org/abs/2312.03732>.
- Edward Turner, Anna Soligo, Mia Taylor, Senthoooran Rajamanoharan, and Neel Nanda. Model organisms for emergent misalignment. *arXiv preprint arXiv:2506.11613*, 2025. URL <https://arxiv.org/abs/2506.11613>.

Appendix A. Training Hyperparameters

Parameter	Value
Base model	unsloth/Qwen2.5-14B-Instruct
Quantisation	4-bit
LoRA rank r	32
LoRA α	64
RSLoRA	yes
LoRA dropout	0.0
Target modules	q, k, v, o, gate, up, down projections
Epochs	1
Effective batch size	16 (per-device 2, grad. accum. 8)
Learning rate	10^{-5} (linear, 5 warmup steps)
Optimiser	AdamW 8-bit
Weight decay	0.01
Max sequence length	2048
Seed	0
Hardware	1 GPU, 64 GB RAM, 6 h (SLURM student)

Table 2: Full training hyperparameters. All three fine-tuned models used the same configuration, only the training dataset differed.

Appendix B. Evaluation Question Set

The test set contains 20 questions across five categories (50 samples each):

- **knowledge:** Quantum mechanics, French Revolution, immune system, seasons, photosynthesis
- **reasoning:** 3-gallon/5-gallon jug puzzle, pros and cons of WFH, explaining inflation to a child
- **writing:** poem about autumn, thank-you email, Crime and Punishment summary, French translation
- **roleplay:** cheering up someone having a bad day, book recommendation, lazy Sunday activities, exam stress
- **practical:** cooking rice, learning programming, tidying before guests, productivity while working remotely

Questions differ from the "irst-plot" questions of Turner et al. (2025), which were designed to maximally elicit EM. Ours cover everyday topics.